University of Dhaka

Department of Biomedical Physics and Technology

BMPT 514

Biomedical Instrumentation Laboratory

Experiment No. 01

Study the different stages of an ECG acquisition circuit and the measurement of limb lead ECG.

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Experiment No. 01

1.1 Name of the Experiment

Study the Different Stages of an ECG Acquisition Circuit and the Measurement of Limb Lead ECG

1.2 Objectives

- To get acquainted with the circuit diagram and data acquisition process in the ECG trainer kit.
- To observe the trace of the six frontal plane ECG.

1.3 Theory

Electrocardiography (ECG) is the process of recording and visualizing the electrical activity of the heart over time. These electrical activities are generated by the depolarization and repolarization of cardiac muscle fibers and can be detected using electrodes placed on the surface of the body.

The rhythmic contraction and relaxation of the heart are controlled by specialized conductive tissues including:

- Sinoatrial (SA) Node
- Atrioventricular (AV) Node
- His-Purkinje System

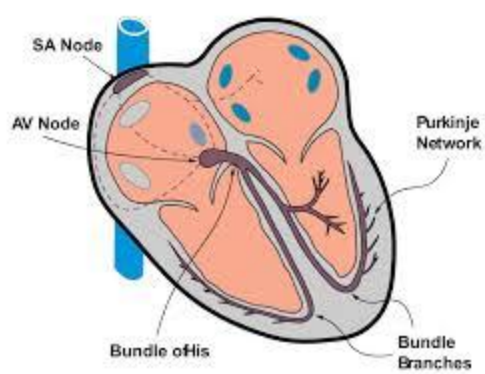


Fig. 01: Cardiac nodes

The electrical activity of the heart can be represented as an electrical dipole. The direction and magnitude of this dipole change continuously throughout the cardiac cycle.

To describe cardiac electrical activity in the frontal plane, Einthoven introduced three standard limb leads forming the Einthoven Triangle.

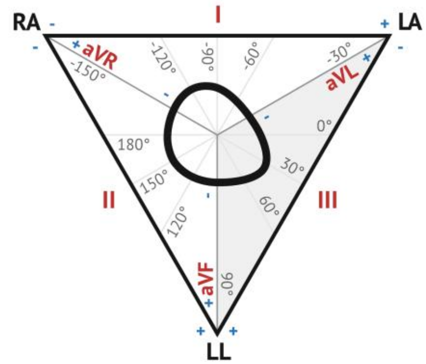


Fig. 02: Einthoven Triangle

Standard Limb Leads

- **Lead I:** Left Arm (LA) – Right Arm (RA)
- **Lead II:** Left Leg (LL) – Right Arm (RA)
- **Lead III:** Left Leg (LL) – Left Arm (LA)

Augmented Limb Leads

Three additional frontal plane leads are derived using Goldberger's central terminal:

- **aVR = RA – (LA + LL)/2**
- **aVL = LA – (RA + LL)/2**
- **aVF = LL – (RA + LA)/2**

The ECG acquisition system employs an instrumentation amplifier and a Right Leg Driven (RLD) circuit. The RLD circuit feeds back the average common-mode signal to the body through the right leg electrode, thereby reducing power-line interference and improving signal quality.

Module Diagram

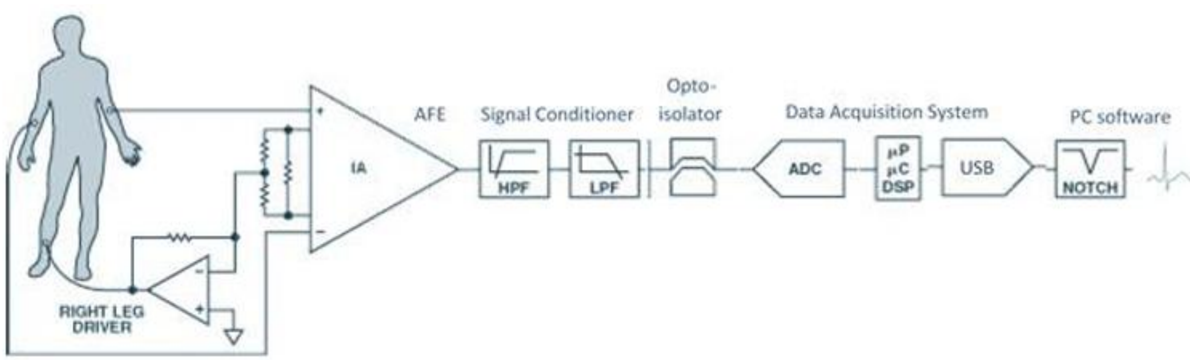


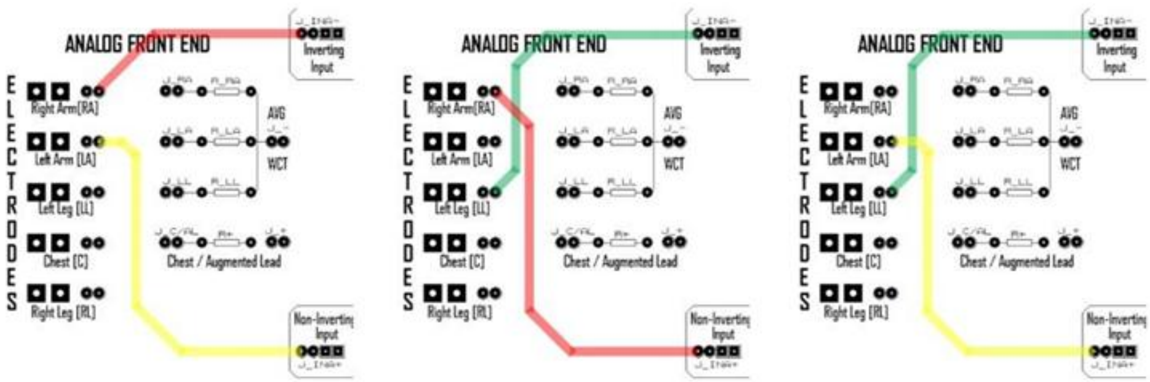
Fig. 03: Block diagram for ECG trainer module

1.4 Equipment

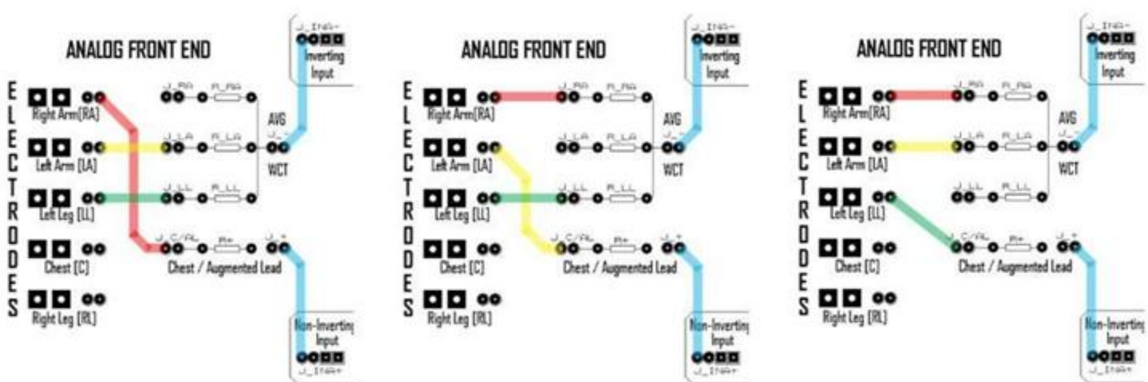
- ECG trainer module
- Computer
- Multimeter

1.5 Setup

A full layout of the experimental setup is presented in Fig. 04.



Lead – I configuration Lead – II configuration Lead – III configuration.



Lead aVR configuration , Lead aVL configuration, Lead aVF configuration.

Fig. 04: Connection for various leads in the ECG Trainer module

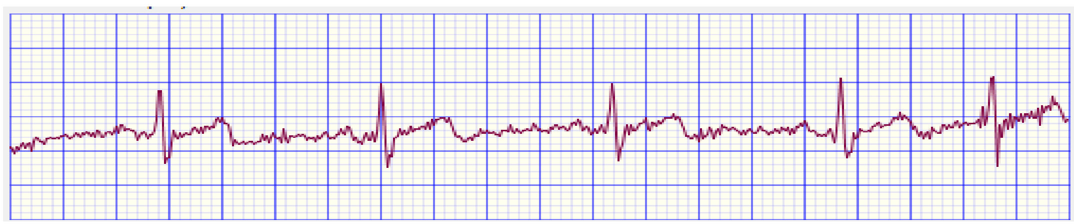
1.6 Data Table

Horizontal scaling: 25 mm/sec
Vertical scaling: 10 mm/mV
Time ECG taken for is: 4 second

Specimen	No of peaks	Duration (sec)	Heart beat rate (bpm)
1	5	4	75

1.7 Calculation

ECG plot of specimen 1 Lead II with respect to RLD: (amplitude vs time in second)

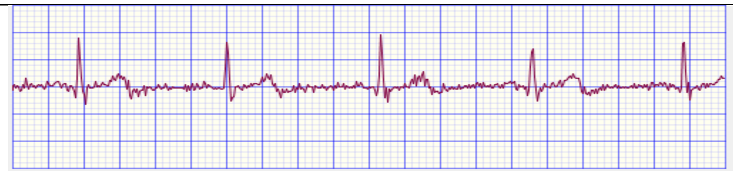
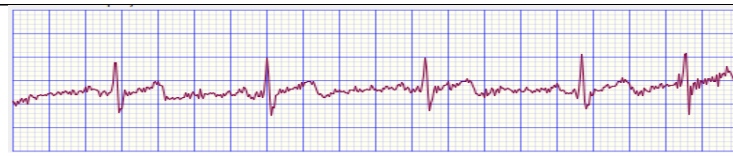
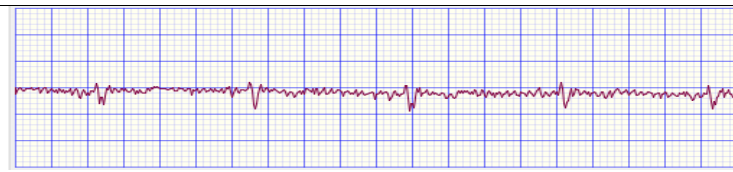
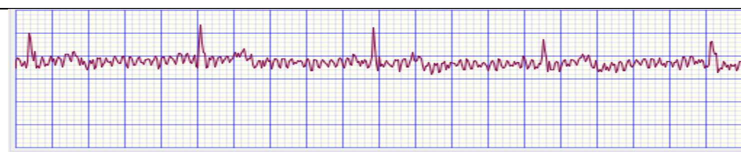
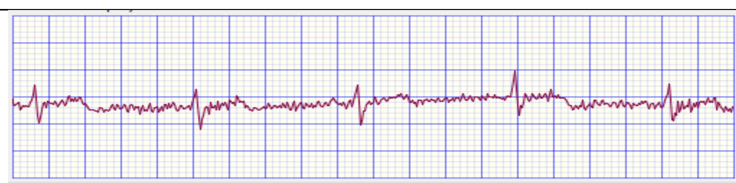


We have taken the ECG of 4 second.

So, heart beat rate of specimen 2 is: $60 \times 5/4 = 75$ bpm.

1.8 ECG Plot

Specimen 1

Lead	w.r.t	ECG Plot
LedI	RLD	
Lead II	RLD	
Lead III	RLD	
Lead aVL	RLD	
Lead aVR	RLD	
Lead aVF	RLD	

1.9 Result

Thus from this experiment, we got the plot of ECG of different leads the specimen.

From the plots:

- Success of ECG system to capture the electrical cardiac activity
- Heart beat rate of specimen: 75 bpm.
- These values are approximate, not exact.

1.10 Discussion

In this experiment, the different stages of an ECG acquisition circuit were observed using the ECG trainer module. The system successfully recorded the electrical activity of the heart through limb electrodes and displayed the ECG waveforms on the computer. The repeated sharp peaks in the ECG traces represent the cardiac cycles, mainly the R-peaks of the QRS complex, which are useful for calculating the heart rate.

From the recorded data, 5 peaks were observed in a 4-second ECG recording. Therefore, the heart rate was calculated as approximately 75 bpm. This value is reasonable for a normal resting condition, but it should be considered an approximate result because the recording duration was short and small counting errors can affect the final value.

The ECG patterns were different in different leads because each lead views the heart's electrical activity from a different direction. Some leads showed clearer and higher peaks, while others had lower amplitude and more noise. These differences may be caused by electrode placement, skin contact, body movement, or external electrical interference. Overall, the experiment helped to understand how an ECG acquisition circuit works and how limb-lead ECG signals can be measured and interpreted.